Land at South Marston and Rowborough
Swindon

Travel Plan Framework

Swindon EV Consortium

Brookbanks

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Executive Summary

Introduction

Brookbanks Consultants Limited (BCL) is appointed by Hallam Land Management, Taylor Wimpey and Hannick Homes to prepare a Travel Plan Framework (TPF) for the planning application for the first phase of the Swindon Eastern Villages (EV) development.

This document investigates different methods of transport that can be used to travel to and from Swindon EV and sets targets to encourage a shift away from use of the private car as the main means of accessing the site.

The approach of the TPF is based upon the principle of identifying appropriate demand management measures that will influence travel behaviour of the new residents and employees of the commercial uses of a particular development proposal.

It is essential that a staged approach of measures and strategies are implemented over time to suit the Swindon EV development phasing and as the level of travel demand increases. This will lead to an effective use of resources and enable measures to be reviewed and adapted when necessary.

The primary goal of the TPF is to minimise the number of car trips to, from and within the site and to promote the use of alternative sustainable modes of transport. The objectives provide the TPF with focus actions on achieving specific outcomes related to the plan.

The key objectives of a TPF include:

- reducing the need to travel by ensuring information is available for the range of facilities available
- reducing the level of car trips at the proposed development
- promoting healthy lifestyles and sustainable, vibrant local communities
- encouraging good design and providing safe pedestrian and cycle access to the site

It is proposed that travel planning activities are implemented by the establishment of a development-wide TPF implemented by the Eastern Villages Steering Group (EVSG). The Steering Group management structure is particularly well suited to mixed use, multiple occupier sites that have wide ranging requirements and impacts.

Each land use will be represented in the EVSG which will be led by a site-wide travel coordinator. The Travel Plan Framework will act as an overarching document that will guide the production of the subsequent individual travel plans.

The key objective of the TPF is to minimise the number of single occupancy car trips to, from and within the site and to advise all occupiers of the site of alternative modes of travel. A package of measures are presented in this document to meet this key objective and so encourage the future residents and employees on site to either to use alternative modes of travel to the single occupancy vehicle, or to car share, and to improve the local environment and health of all site occupiers.

In order to achieve the objectives set out in this Travel Plan Framework, it is essential that for a development of this size, that the measures and methods used are implemented using an approach that covers the various stages outlined and divided into key phases:-

- Pre-occupation
- During occupation

The figure below outlines the measures to be implemented to achieve the objectives, and who would be responsible for their implementation and monitoring.

Measure	*Responsible Person*	*Timescale*
Pre-occupation Stage		
A fully permeable layout through the site for pedestrians and cyclists reflecting desire lines	Developer	Included with the approved site layout
Parking in line with current guidance	Developer	Included with the approved site layout
On-site streets and junctions designed to the standards outlined in the Department for Transport's Manual for Streets	Developer	Included with the approved site layout
Shared surfaces on residential loop lanes to promote pedestrian movements, creating a 'Home Zone' style street environment	Developer	Included with the approved site layout
Walking and cycling on site linked to existing external routes.	Developer	Included with the approved site layout
Careful street design to restrict vehicular speeds to 20mph or less within home zones	Developer	Included with the approved site layout
Secure and conveniently located cycle storage provision	Developer	Included with the approved site layout
Appointment of Site Wide Travel Plan Coordinator	Developer	Prior to first occupation
Off site Infrastructure Improvements	Developer	To be delivered as per the S106
Development of Welcome Packs	Travel Plan Coordinator	Prior to first occupation
Establish steering group	Developer	Prior to first occupation
During Occupation		
Improvements to public transport routes Establish TMA	Developer Developer	On first occupation On first occupation
Distribution of Welcome Packs	Travel Plan Coordinator	On first occupation
Sales staff training	Travel Plan Coordinator	On establishment of sales office
Establish Bus Buddy databases	Travel Plan Coordinator	Prior to first occupation
Establish Cycle user group	Travel Plan Coordinator	Prior to first occupation
Provision of regular travel information	Travel Plan Coordinator	As and when necessary
Establish car club	Travel Plan Coordinator	On 500th occupation
Promotion and awareness raising packages	Travel Plan Coordinator	On an annual basis
Establish car share database	Travel Plan Coordinator	On first occupation
Establish dedicated web site	Travel Plan Coordinator	On first occupation
Coordination of appropriate reviews and monitoring of the success of the TP.	Travel Plan Coordinator	On occupation of 800th dwelling
Initial Travel Pattern Survey	Travel Plan Coordinator	On occupation of 800th dwelling
Monitoring surveys	Travel Plan Coordinator	On every 800th completion intervals
Provision of regular updated travel information	Travel Plan Coordinator	As and when necessary
Updates to the travel plan	Travel Plan Coordinator	annually
Coordination with Local Authority	Travel Plan Coordinator/CCC	Annually

1 Introduction

Introduction

1.1 Brookbanks Consultants Limited (BCL) is appointed by Hallam Land Management, Taylor Wimpey and Hannick Homes to prepare a Travel Plan Framework (TPF) for the planning application for the first phase of the Swindon Eastern Villages (EV) development, which is envisaged to comprise 2,380 dwellings.

1.2 The complete development is envisaged to comprise circa 8,000 residential units, employment land uses, local centres and primary and secondary schools. However, this TPF primarily considers the anticipated planning application submission for the first phase of the development that deliver circa 2,380 dwellings. As further planning applications are submitted for subsequent phases, this Framework will be updated in line with the specific details of the developments.

1.3 This TPF establishes the initial framework of travel initiatives to be pursued by the developer. It forms a strategy for influencing change in residents and employees travel patterns with the intention of providing a disincentive for non-essential car use to the site. This document will provide a framework for the development of the residential travel plans that will be submitted as part of the reserved matters for phase I.

1.4 This document investigates different methods of transport that can be used to travel to and from Swindon EV and sets targets to encourage a shift away from use of the private car as the main means of accessing the site.

Proposed Development

1.5 Phase 1 of the Eastern Villages proposed development site is located approximately 6km east of Swindon town centre, 1km to the east of the A419 Cirencester to M4 trunk road and on the north side of the A420 Swindon to Oxford principal road.

1.6 Phase 1 of the development is indicate in Figure 1a.

1.7 The proposed first phase of the wider Eastern Village development will include

- 2,380 residential units providing a mixture of type and tenure to meet future demand for open market and social housing
- Primary school
- Local centre providing a mixed range of local facilities
- Low scale Employment provision
- Open space provision

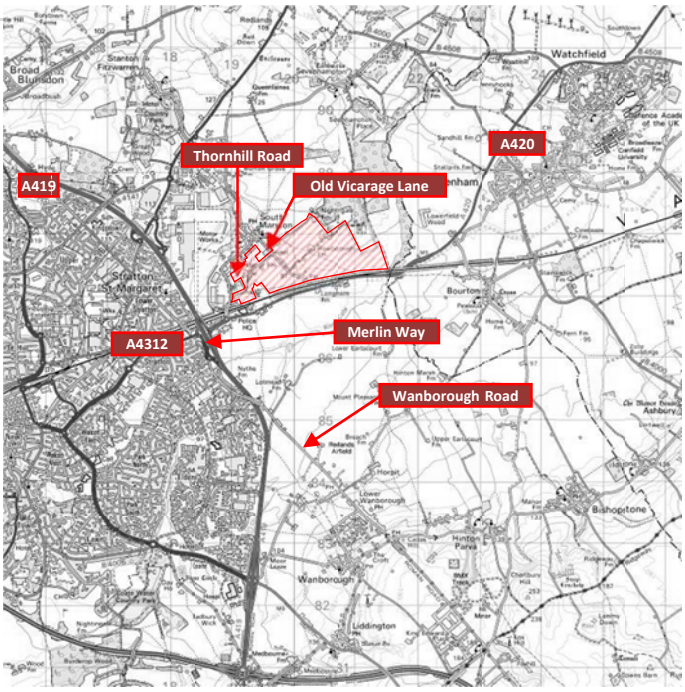


Figure 1a: Location Plan

Travel Plan Context

1.8 A Transport Assessment (TA) is being prepared by BCL in support of the initial phase of the Swindon EV planning application. Part of the proposed transport strategy for the development is the implementation of a Travel Demand Management Strategy to reduce single occupancy car trips associated with Swindon EV. A key part of the strategy is the preparation, management, and review of a Travel Plan Framework.

1.9 This Travel Plan Framework has been prepared with specific reference to national guidance by the Department for Transport, specifically 'Making Residential Travel Plans Work: Good Practice Guidelines for New Development' (DFT, 2005), 'Making Residential Travel Plans Work' (DFT, 2007), 'Smarter Choices – Changing the Way We Travel' (DfT, 2004), Good Practice Guidelines; Delivering Travel Plans through the Planning Process (DfT April 2009) and The Essential Guide to Travel Planning (DfT March 2008).

1.10 The preparation of the residential travel plans should take into account local guidance available from Swindon Council, with details provided at www.swindontravelchoices.org.uk. The employment travel plans that are to be completed as part of the later stages of development will take into account Swindon's Workplace Initiative For Transport (SWIFT). This provides a forum for employers to discuss transport issues and provides a range of supporting travel plan materials.

Travel Plan Overarching Objectives

1.11 The key objectives of the TP are to:

- Maximise modal shift to sustainable travel
- Ensure development and implementation of a series of site specific smarter modal choice measures that support and promote sustainable travel
- Ensure an ongoing process for the coordination, monitoring and management of the implementation of the measures and to review changes to ensure achievement of modal shift.

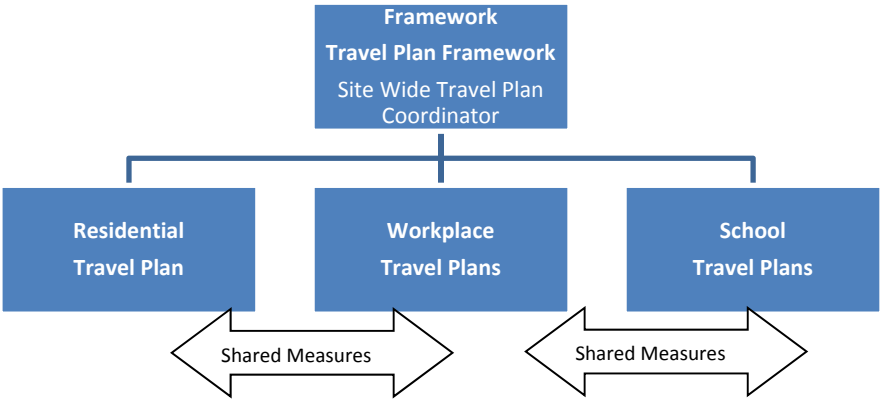
1.12 This TPF is a working document and achievement of the objectives will require participation from all the future residents/occupiers. The TPF is a strong indication of the commitment of the developers to the achievement of sustainable travel targets.

1.13 Given the complexity and scale of development proposals, it is essential that this TPF is read and assessed in conjunction with the Transport Assessment submitted with the Planning Application.

1.14 It is proposed that travel planning activities are implemented by the establishment of a Eastern Villages wide TPF implemented by the Eastern Villages Steering Group (EVSG). The Steering Group management structure is particularly well suited to mixed use, multiple occupier sites that have wide ranging requirements and impacts.

1.15 Each land use in due course will be represented in the EVSG which will be led by a site-wide travel plan coordinator. The Travel Plan Framework will act as an overarching document that will guide the production of the subsequent individual travel plans for phases of development including north of the A520, the current application.

1.16 The flow chart below demonstrates this approach.



1.17 Due to the scale of employment use within the second phase of development south of the A420, it is considered that a Transport Management Association (TMA) in due course could be established. The Transport Management Association will provide a means of forming and maintaining effective partnerships across the development and it will be a central element of the Travel Plan Implementation Strategy.

1.18 The TMA is initially funded by the developer and will also obtain revenue from employers based on site. Following full build-out of the employment element during subsequent development phases of the development, the TMA is designed to be self funding in order for the Travel Plan to continue operating. Further details should be provided on the TMA as part of any subsequent planning application for development south of the A420.

1.19 The Employment Travel Plan and TMA will be managed by the Employment Travel Plan Coordinator, whilst the residential Travel Plan will be managed by the Residential Travel Plan Coordinator.

1.20 The site-wide travel coordinator will oversee all the Travel Plans produced by the individual travel plan coordinators or each element of the development. Ultimately, this will ensure an integrated and consistent approach between the plans.

1.21 The Travel Plan Framework will be secured by Section 106 agreements associated with the application for the Eastern Village to ensure implementation. This document is the Travel plan Framework, but also describes a number of specific proposals and approaches in relation to the current application north of the A420.

2 Travel Plan Principles

2.1 The TPF sets out a holistic package of measures designed predominately to reduce single occupancy car use associated with the proposed development, by supporting and providing alternative forms of transport and reducing the need to travel. The measures will be integrated into the design, marketing, and occupation of the site.

2.2 A Sustainable Transport Strategy has been developed for the Swindon EV to promote all sustainable modes of transport, this includes public transport, walk and cycle modes.

Key Principles

2.3 The approach of the TPF is based upon the principle of identifying appropriate demand management measures that will influence the travel behavior of the new residents. These measures will then be carried forward into the residential and workplace travel plans.

2.4 It is essential that a staged approach of measures and strategies are implemented over time to suit the Swindon EV development phasing and as the level of travel demand increases. This will lead to an effective use of resources and enable measures to be reviewed and adapted when necessary.

Key Benefits

2.5 The benefits of a Travel Plan include:

- A focused approach to influence travel behaviour and manage demand for travel to the site
- A reduction in the need to travel and where appropriate, the need for journeys to be single occupancy car journeys
- The introduction of safe and viable alternatives to single occupancy car travel, improving both the availability and choice of travel mode to the site and reducing the number of vehicle trips
- Increased awareness of the potential for and advantages of travelling by sustainable modes of travel, including walking, cycling, public transport and car sharing
- Increased awareness of the social, environmental and economic costs of individual travel choices
- A positive change in attitudes towards the use of alternative travel modes
- Provision of practical initiatives based on regular appraisal of staff travel patterns
- Improved accessibility for all
- Improved safety and security for people travelling to and from the site

Key Guidance

2.6 There is an extensive library of documents that illustrates best practice guidance and evidence relating to travel plans. Key guidance documents include:

- Guidance on Transport Assessment (DfT, 2007)
- Smarter Choices - Changing the way we travel (June 2005)
- Making Residential Travel Plans Work: Good Practice Guidelines for New Development (DfT, 2005);
- Using the planning process to secure travel plans. Best practice guidance for local authorities, developers and occupiers (DfT 2002);
- Transport Energy Best Practice – A Travel Plan Resource Pack For Employers (DfT, 2002)
- Making Travel Plans Work (DfT, 2002)

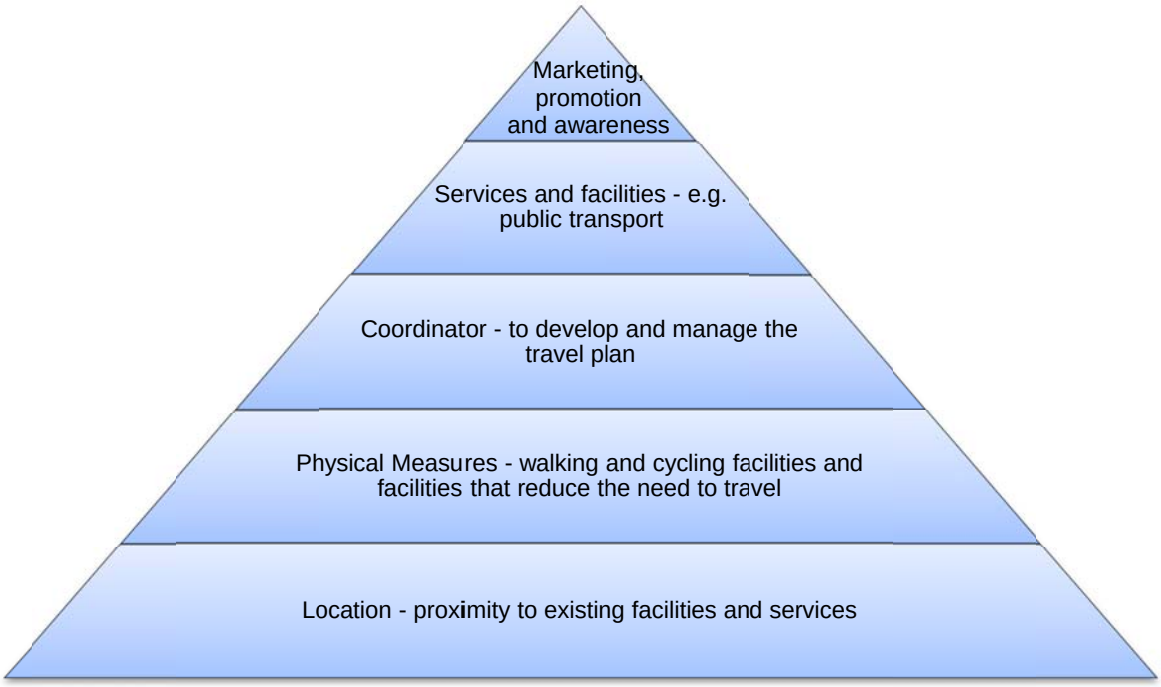
Types of Travel Plan and Role of this Document

2.7 Travel Plans prepared in advance of the occupation of a site, whether for residential, employment or school use, can only offer an overall strategy for the adoption of sustainable transport measures. Once a site has achieved a sufficient level of occupancy, then the users must have the opportunity to influence and amend the Travel Plan to reflect their specific needs, while meeting the plan's original objectives and associated planning commitments.

2.8 This is particularly an issue for residential developments, where it can be problematic to maintain momentum and achieve "ownership" over a long period. The developer's approach has been to embed measures into the design of Swindon EV from the start of the process, through ensuring good physical and sustainable infrastructure provision and plans for long-term management and monitoring, as discussed and outlined in this Travel Plan Framework.

2.9 This Travel Plan Framework is therefore, a first issue of a working document that establishes a framework for the production of the subsequent travel plans, as it requires participation from the residents that will occupy Swindon EV. The document however, provides a strong indication of the objectives, priorities and commitment of the developer to achieve high levels of sustainable transport use.

2.10 To ensure the success of the Travel Plan it is important to establish a clear strategy. The DfT travel plan strategy referred to in the Good Practice Guidelines: Delivering Travel Plans through the planning Process demonstrates the basis of sound travel planning, as indicated below.



3 Eastern Villages Travel Planning Overview

3.1 The TPF will not focus on one particular land use, rather provide an overview. There will be specific tailored measures to cover the greatest possible range of trips including residential, community and commercial. It is therefore proposed that individual residential and workplace Travel Plans are developed for the site at the appropriate time.

3.2 This overall TPF represents the travel planning strategy for the development and will include a range of measures and initiatives common to a range of occupiers and activities.

3.3 The TPF will provide a guide to the development of specific travel plans where these are required.

3.4 This TPF outlines a range of measures that will be commenced and implemented during the early phases of the development. This approach encourages people to adopt sustainable travel patterns from the outset.

4 Existing Accessibility and Future Accessibility

4.1 This section provides details of the site location and the existing transport modes. Detailed information can be found within the Swindon EV Transport Assessment.

4.2 The site location in relation to its accessibility to existing transport infrastructure and potential increased accessibility lends to the success of the Travel Plan and achieving modal shift by altering travel behavior.

Site Location

4.3 The proposed first phase of development lies circa 6km east of Swindon town centre, to the north of the A420 and railway line. The first phase development area is in two parts, one to the west of Old Vicarage Lane and one to the east. The western part of the site is bound by Old Vicarage Lane, Thornhill Road, South Marston village, and the railway. The eastern part of the site is bound by Old Vicarage Lane, Nightingale Lane and by agricultural land.

4.4 The majority of the land is currently undeveloped and not thought to have been subject to historic development other than farms. However, small developments including a combination of dwellings, industrial units and hotels are located adjacent to the site. The site location and boundary are shown indicatively on Figure 4a, below.

Figure 4a: Site location

Existing accessibility

4.5 Good accessibility from the site to key destinations by modes other than car encourages more sustainable modes of travel which assists in reducing single car occupancy. In particular, the location of employment, retail, education and community facilities are some of the most common forms of trip generators.

Existing Road Network

4.6 The location of the site in relation to the local road network is indicated in Figure 4b.

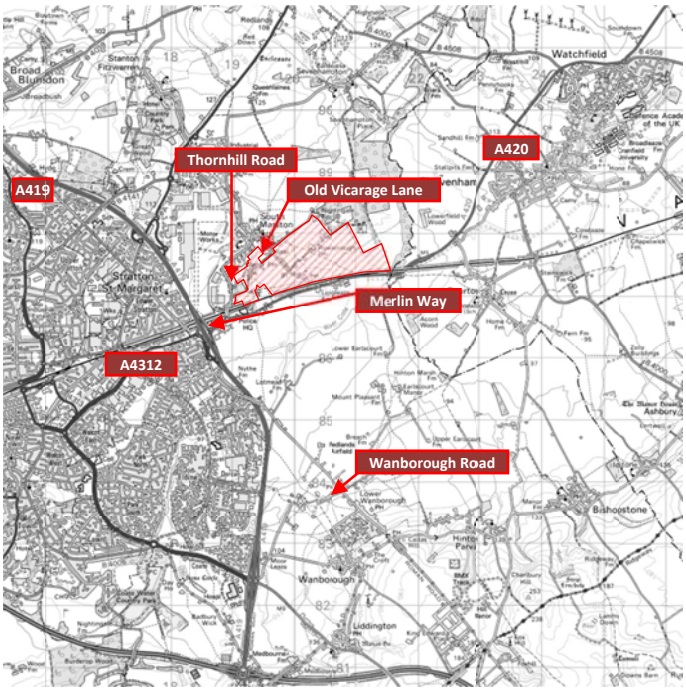


Figure 4b: Site Location in relation to the local road network

4.7 The majority of the road network adjacent to the site is classified as part of the Local Road Network (LRN), managed by SBC, with the A419 and the M4 Motorway managed by the HA. The roads managed by the HA are typically identified as being a part of the Strategic Road Network (SRN).

4.8 The A420 is located to the south of the site and forms an important corridor within the LRN. The A420 commences from Swindon and heads generally in a north easterly direction extending into Oxford. The A420 is predominantly single carriageway road which is subject to 50mph speed limit along much of the length. There are sections of dual carriageway and the section approaching Swindon is subject to a 50mph speed limit.

4.9 There are a several junctions along the length of the A420 which are predominantly at-grade. The junctions along the A420 that are within the hinterland to the site are:

- A420 / Old Vicarage Lane
- A420 / Police HQ
- A420 / Gable Cross
- A420 / A419

4.10 Old Vicarage Lane forms a priority T-junction with the A420 with a ghost island right turn bay for vehicles travelling along the A420. From the junction with the A420, Old Vicarage Lane continues in a north westerly direction towards the village of South Marston. Old Vicarage Lane is rural in character and does not provide any dedicated facilities for pedestrian or

cyclists. Circa 100m north of the A420, Old Vicarage Lane passes under the Swindon to Paddington rail line, as indicated below. The carriageway beneath the underbridge is of reduced cross sectional width.

Figure 4c: Southerly view along Old Vicarage Lane

4.11 To the north, Old Vicarage Lane forms simple priority junctions with Nightingale Lane and Rowborough Lane. These are minor roads which provide access to a small number of existing residential properties.

4.12 Circa 1.4km to the west of the junction with Old Vicarage Lane, the A420 provides a signalised access junction to the Wiltshire Police headquarters. There is no facility to undertake a right turn from the A420 into the police headquarters, so vehicles needing to access the facility from an easterly direction must continue through this junction toward the Gablecross roundabout and undertake a U-turn. Heading in a westerly direction from this junction, there is an off road segregated walking and cycling route.

4.13 The Gablecross roundabout lies circa 200m to the west of the Police HQ signals and is a five arm roundabout. The arms of the roundabout serve Sainsbury's supermarket to the south, Thornhill Road and the St Margaret's Retail Park to the north with the remaining arm being exit only from the retail park. The off-road pedestrian/cycle route continues westerly through Gablecross junction. There is an unsignalised pedestrian crossing across the Thornhill Road arm with a signalised pedestrian crossing to the west of the junction to provide a safe route into the supermarket.

4.14 Thornhill Road heads in a generally northerly direction to access the South Marston village. Thornhill Road is rural in nature and does not provide for dedicated facilities for pedestrian or cyclists within the highway over much of the length, as indicated below. There is however a pedestrian and cycle route alongside Thornhill Road between the Gablecross roundabout and the Keypoint roundabout and there is a segregated pedestrian and cycle route to the west of Thornhill Road between the Keypoint roundabout and the start of south Marston village. There is traffic calming (single file traffic) at the entry and exit of the village which attempts to control vehicle speeds into South Marston

4.15 Thornhill Road continues into the village of South Marston where there are sections of a discontinuous footway. In the centre of South Marston, Thornhill Road forms a priority T junction with Old Vicarage Lane, with this junction known locally as Pound Corner. The village is accessed by three roads (Highworth Road, Thornhill Road and Old Vicarage Lane) that link to the north with the A361 and to the south with the A420 and onward to the A419 trunk road and M4. These roads are considered to be potential "rat runs" between the strategic roads and to the major employment areas adjacent

to the village. The alignment and relative narrowness of these roads create points of potential pedestrian and vehicular conflict, including Pound Corner.

Figure 4d: Northerly view along Thornhill Road

4.16 Circa 300m to the west of the Gablecross junction lies a grade separated junction between the A419 and the A420, known as the White Hart interchange. The interchange has two parts that are connected by part of Merlin Way. The northern part of the interchange is a grade separated roundabout with the A419 main carriageway passing over the roundabout. The roundabout connects the A420, the A419 southbound slip roads on the east side of the roundabout together with Oxford Road and Ermine Street to the west and Merlin Way to the south. The northbound A419 on and off slips have a roundabout junction with Merlin Way that is located 400m to the south of the White Hart roundabout junction and form the southern part of the interchange . Any vehicles going to the north on the A419 from the A420 have to travel southwards along Merlin Way before travelling north along the A419 and as such the present arrangement is counter intuitive. Oxford Road continues from the White Hart interchange westerly towards Swindon town centre.

4.17 The White Hart interchange incorporates a pedestrians and cyclist bridge linking the A420 movements east-west either side of the A419.

4.18 A Toys-R-Us store is located to the south east of the A419. Inbound access to the store is taken direct from the A419 southbound on slip. Outbound trips joining the A420.

4.19 Merlin Way continues south from the White Hart interchange and becomes Wanborough Road which crosses over the A419 via a single carriageway road bridge 1.5km south of White Hart Roundabout. Wanborough Road links with Wanborough and Upper Wanborough, and has one lane per direction.

4.20 Continuing from the White Hart interchange, Oxford Road (A4312) provides a westerly link into Swindon town centre. Oxford Road is a single carriageway road with a footway to the north and a footway/cycle way to the south. Oxford Road is a residential road allowing direct access to individual properties. Oxford Road has a signalised pedestrian crossing near Wanborough Road, a signalised junction with Nythe Road and a roundabout junction with Dorcan Way, with this junction being identified as the Green Bridge roundabout, being adjacent to the Green Bridge Retail Park.

4.21 From the White Hart interchange the A419 heads in north westerly and southerly directions. To the north west the road heads toward Cirencester passing through further junctions identified locally as the Honda and Rat Trap roundabouts. To the south, the road passes through the Commonhead junction to link with the M4 at Junction 14. The A419 takes the form of a lit dual-carriageway which is subject to a national speed limit of 70mph. There are no pedestrian or cyclist facilities along the A419.

4.22 The M4 motorway caters for east – west movements and provides the major ink between South West England / Wales to the west and Reading / London to the east. The grade separated junction with the M4 is signalised. The roundabout is lit and there are no pedestrian or cyclist facilities at the junction.

Road Based Public Transport

4.23 Bus services 65 and 66 operate along the routes close to the site as indicated below.

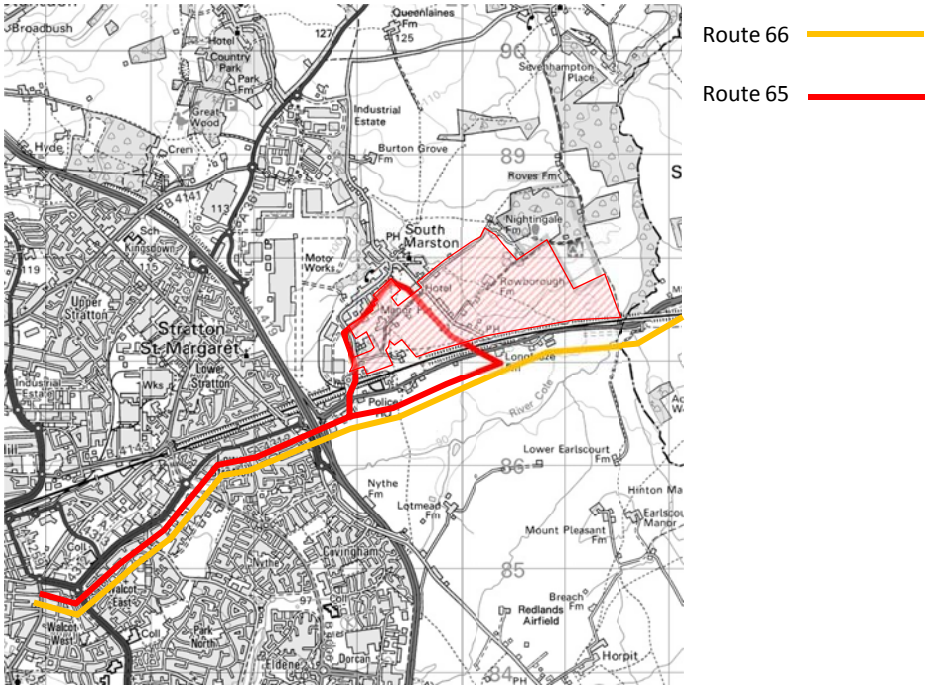


Figure 4e: Bus Routes operating close to the EV

4.42 The bus service 65 is a route managed by Stagecoach and operates between Swindon and Faringdon, Monday to Saturday. The first bus leaves Swindon Bus Station at 10:50 calling at South Marston at 11:03. For trips towards Swindon, the first bus calls at South Marston at 10:26 with a journey time of 15 minutes. This service operates typically four times daily.

4.43 The Route 66 is managed by Stagecoach and operates along the A420 between Swindon and Faringdon, Monday to Sunday. The first bus leaves Swindon at 05:40 and typically operates on a half hourly frequency. The journey time to Oxford is 1 hour and 15 minutes.

4.44 The bus station provides waiting areas, toilets and is located circa 750 metres to the south of Swindon train station.

Rail Based Public Transport

4.45 The closest train station is located in central Swindon, circa 6km to the west of the proposed site. The train station, managed by First Great Western, provides a range of facilities including:

- 578 space car park open 24 hours
- Ticket office open 7 days of the week
- Self-service ticket office
- Manned help desk
- Cash machine
- Pay phones
- Post box
- Public WiFi
- Refreshments
- Shops
- Toilets
- Waiting rooms

4.46 Swindon rail station provides three routes per hour to Didcot with a journey time of circa 17 minutes. From Didcot, onward travel to Oxford is possible. Additional routes to London Paddington are provided three times per hour with a journey time of 63 minutes.

4.47 Trips from Swindon westwards to Bristol are supported, with two trips per hour each to Temple Meads (45 minute journey time) and to Bristol Parkway (Journey time 27 minutes).

Walking and Cycling

4.48 The development site does not have any significant traffic generators on site such that there has been no requirement for significant footway provision on site or to access the site.

4.49 The primary roads that do provide footways have been identified earlier in this section.

4.50 Swindon has a cycling network consisting of a mixture of both on and off road routes that also caters for walking trips. As indicated earlier an off road cycling route commences from the police headquarters and continues westerly towards the town centre. A cycle way extends northwards from the junction with Gablecross to link South Marston. There are additional routes to the east of the site.

4.51 The figure below identifies the location of the cycleways closest to the site.

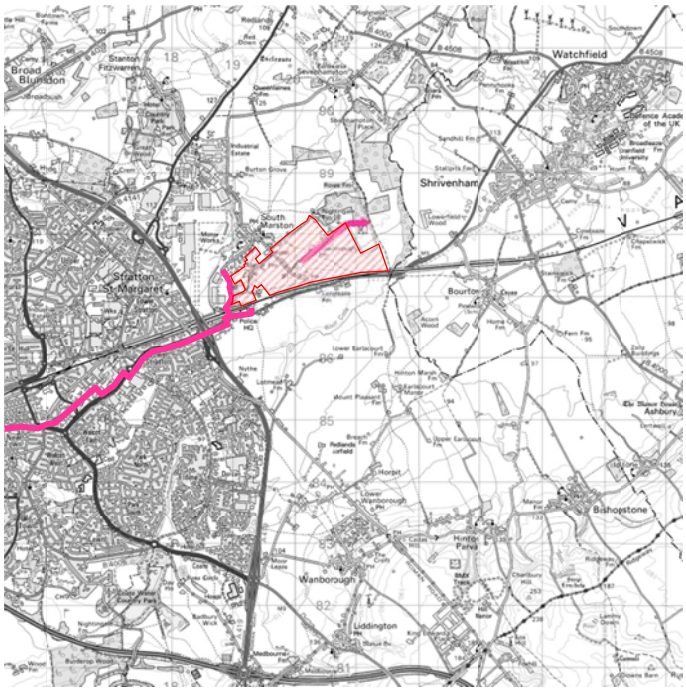


Figure 4f: Cycleways closest to site.

4.52 The desire line from the site is predicted to be westerly towards the supermarket and the town centre. As indicated above there is walking and cycling link along Thornhill Road. This provides a continuous link into the supermarket and towards the White Hart interchange. The interchange provides a bridge for onward travel towards the town centre.

Public Rights of Way

4.53 Public Rights of Way (PRoW) are classified as highways and as such are protected routes. The 1949 National Parks and Access to the Countryside Act placed a duty on every County Council in England and Wales to draw up and publish a definitive map and statement of PRoW in their area.

4.54 The Definitive Map is the legal record of the location and status of PRoW. The statement is a description of the PRoW shown on the definitive map.

4.55 There are four classifications of PRoW:

- footpaths - by foot only
- bridleways - by foot, horse or bike
- restricted byways - by any form of transport that doesn't have a motor
- byways open to all traffic - let you travel by any form of transport, including cars

4.56 The figure below highlights the PRoW that are closest to the site. This illustrates that there is a network of bridleways and footpaths that cross the site that connect with bridleways that penetrate other roads in the vicinity of the site. In particular there are existing strong links between South Marston and the A420 for pedestrians and cyclists to Gablecross and for pedestrians across the railway at underbridges further to the east near the Police HQ and to the west and east of Old Vicarage Lane.

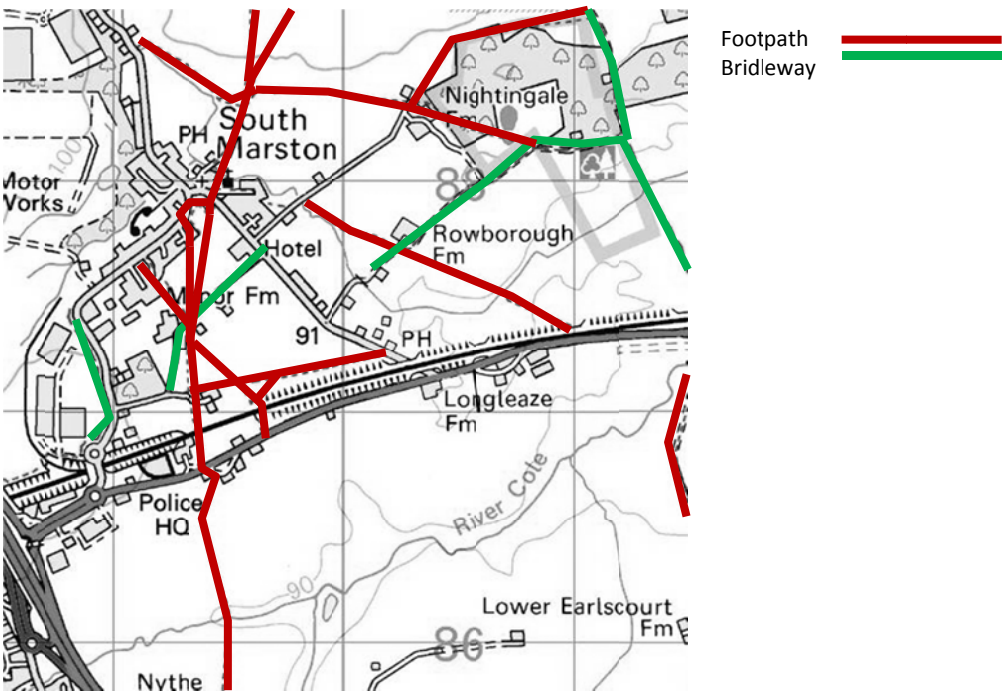


Figure 4g: Public Rights of Way

Future accessibility - Phase One

4.57 In order to improve accessibility to public transport, a Public Transport Strategy will be developed in discussion with local transport operators and Swindon Council.

Public Transport

4.58 To maximise the opportunities to travel by public transport it is proposed to improve the current routes that operate in Swindon. The options to deliver public transport enhancements have been discussed with SBC. Currently there are only two routes operated by Stagecoach that operate adjacent to the site boundary.

4.59 Through discussions with Stagecoach, various public transport options were investigated. This included rerouting of the existing services through the development.

4.60 The development has been assumed to commence from Thornhill Road and proceed in an easterly direction. Therefore a link through the development into Old Vicarage Lane is unlikely to be delivered at the outset. Through discussions with Stagecoach, Route 66 should only be routed through the development once the link with Old Vicarage Lane is in place.

4.61 Therefore alternative routes were investigated to serve the initial phase 1a. Route 72 was identified as a viable alternative. This route currently serves Wrougton to the south of Swindon and terminates at the bus station. This route could be extended to serve the EV Monday to Saturday at half hourly intervals with the first call at 0624 with the last stop at 2054.

4.62 The development will deliver a link through to Old Vicarage Lane, at which point the 66 will be diverted into the site. The 72 will revert back to the original route unless patronage is sufficient that the route is self funding.

4.63 Route 66 will operate at an improved 20 minute frequency Monday to Saturday and hourly on Sunday.

4.64 The long term viability of any public transport routes is critical. Therefore through the discussions, Stagecoach were requested to estimate the likely level of revenue that could be generated by the development, offset by the likely costs. This indicated that the long term viability of the public transport intervention can be secured.

Walking and Cycling

4.65 Published good practice identifies five main requirements for pedestrian routes. Wherever possible these should be followed when planning for pedestrians within the proposed development:

- Convenience – follow desire lines without any undue deviation from route,
- Connectivity – link multiple origin and destinations,
- Conviviality – be pleasant to use,
- Coherence – be made legible through paving and/or signage,
- Conspicuousness – promote security and safety allowing pedestrians to see and be seen by others

4.66 The 'Guidance for Cycle Audit and Cycle Review' (The Institution of Highways and Transportation, 1998) determines five main requirements for cycle routes. It is highly crucial that these requirements are recognised if the promotion of cycling to the site as a viable and attractive alternative to car use is to be successful:

- Coherence – continuous and to a consistent standard,
- Directness – closely follow desire lines as much as possible,
- Attractiveness – in aesthetic as well as objective terms
- Safety – designed to minimise risks for cyclists and others; and
- Comfort – well maintained smooth dry surfaces, flush kerbs and gentle gradients

4.67 Overall consideration should be given towards the former Commission for Architecture and the Built Environment (CABE) principles of inclusive design, as highlighted below:

- Inclusive - so everyone can use it safely, easily and with dignity.
- Responsive - taking account of what people say they need and want.
- Flexible - so different people can use them in different ways.
- Convenient - so everyone can use them without too much effort or separation.
- Accommodating - for all people, regardless of their age, gender, mobility, ethnicity or circumstances.
- Welcoming - with no disabling barriers that night exclude some people.
- Realistic - offering more than one solution to help balance everyone's needs and recognising that one solution may not work for all

4.68 The strongest desire line for walking and cycling is westbound. The distribution of trips, as indicated later in this report is 74% towards Swindon. Based on the likely trip generation, 788 vehicle trips, based on the modal split identified early, this could resukt in 1134 trips (788/0.695). This in turn could result in circa 110 walking and cycling trips heading towards Swindon.

4.69 The masterplan for the site will include numerous walking and cycling routes within the development to provide a comprehensive route network that will comprise both on and off road paths. Highway crossing points will be designed to cater for all types of pedestrian users with the routes lit where appropriate.

4.70 The walking and cycling paths will connect the individual housing blocks into the main route through the site that will ensure full connectively and route choice throughout the development.

4.71 The onsite network will connect into the external walking and cycling networks that includes the PRoW network. The predominant walking and cycling desire line towards Swindon is to be fully incorporated into the links from the development. Currently there is a high quality walking and cycling link adjacent to the A420 commencing from the Police headquarters junction heading in a westerly direction through the Gablecross junction to connect with the Whitehart interchange. The development will provide a connection into the walking and cycling link adjacent to Thornhill Road, which in turn links with the A420. This will ensure that trips into the supermarket and into Swindon can be carried out.

4.72 It can be concluded that the site will be supported by a range of measures to improve overall accessibility. The potential measures are discussed in later sections of the report but in brief will include improvements to the public transport network, introduce a car share database and contribute towards the establishment of a car club.

Future accessibility

4.73 At the time of writing, it is understood that subsequent phases could deliver an additional 6,000 dwellings together with employment land uses. It is expected that this could generate additional revenue streams which could support the provision of additional buses. At the time of writing, it is difficult to predict the future likely destinations. However it is considered that this provides an opportunity to deliver a comprehensive public transport package.

5 Travel Plan Objectives and Targets

Introduction

5.1 As with all new developments, there are no established patterns of behaviour for residents, employees and visitors or deliveries to the development. Therefore, this document forms a Framework for managing and informing travel arrangements for those future uses. Once the development is occupied and the actual travel patterns and needs are known, more detailed and specific measures, incentives and targets based on recoded travel patterns will be tailored to the development

5.2 The primary goal of the TPF is to minimise the number of car trips to, from and within the site and to promote the use of alternative sustainable modes of transport. The objectives provide the TPF with focus actions on achieving specific outcomes related to the plan.

5.3 The key objectives of a TPF include:

- reducing the need to travel by ensuring information is available for the range of facilities available
- reducing the level of car trips at the proposed development
- promoting healthy lifestyles and sustainable, vibrant local communities
- encouraging good design and providing safe pedestrian and cycle access to the site

Objectives and Targets

5.4 The key objective of the TPF is to minimise the number of single occupancy car trips to, from and within the site and to advise all occupiers of the site of alternative modes of travel. A package of measures are presented in this document to meet this key objective and so encourage the future residents on site to either to use alternative modes of travel to the single occupancy vehicle, or to car share, and to improve the local environment and health of all site occupiers.

5.5 An important feature of the TPF is the appointment of the Site Wide Travel Plan Coordinator (SWTPC) and the individual travel plan coordinators for the new Eastern Villages at the appropriate time to promote alternatives to single occupancy car travel and promote awareness of car sharing and provides information on the incentives.

Outcomes

5.6 It will be important to put measures in place that promote sustainable travel through education and the dissemination of information. Targets will be set to reduce car-based trips over a monitoring period which will include regular travel surveys.

5.7 All the TPCs will encourage and promote the measures set out along with infrastructure changes agreed with the local planning authority to achieve the objective of the TPF. These initiatives will be reviewed with reference to findings from the monitoring travel survey.

5.8 Furthermore, in line with the 'Delivering Travel Plans through the Planning Process-Good Practice Guidelines dated March 2008, the TPF will also seek the following outcomes:

- Reduction in the need for car use, with associated benefits in terms of reduced traffic, congestion, air pollution and accidents
- Reduction in overall need for travel to and from the site
- Generation of only the minimum number of car traffic movements to and from the development
- Improved viability of local public transport services
- Increased usage and safety of neighbourhood cycling and walking routes
- Opportunities provided to build healthy exercise into daily life
- Represent good practice and provide an educational tool to help change perceptions about convenience and benefits of not using the car where alternatives exist
- Increase marketability of the development as more people seek to change their travel behaviour

5.9 The design of the development, and the travel planning process, will aim to reduce overall need to travel, and attempt to challenge choices made by all occupiers on whether to drive given access to a car, how much to drive, and how much to travel by public transport, foot or cycle. Where car travel is necessary, the TPF will aim to minimise single occupancy car trips and promote car sharing. Table 5.1 shows the Objectives and targets.

Objectives	Targets	Indicator
To establish Travel Plan Steering Group	To establish the operational steering group by first occupation	Steering Group set up
To achieve widespread awareness of sustainable travel within the minds of all occupants of the site	To achieve and maintain at least 75% awareness during build out period	Travel Survey Results
To reduce external car driver trips associated with the development on the local and strategic highway network	Reduction of external car driver trips generated by the residential elements	Data from ATC Monitoring and Travel Survey
To contribute to reducing the impact of Swindon EV travel on climate change	Monitor and evaluate, at milestones carbon emissions from local trips	Data from Travel Surveys
To increase the proportion of sustainable travel modes (Walk & Cycle)	To increase walk and cycle trips.	Data from Travel Surveys
To engage with residents on sustainable travel choices	All new households to be offered personal travel planning within 12 months of occupation	TPC

6 Eastern Villages Travel Plan Implementation Strategy

6.1 A responsive, flexible and dynamic Travel Plan implementation process is extremely important for a site that will be developed over an extended period.

6.2 The implementation strategy of the Residential and Workplace Travel Plans remain the responsibility of the developer or occupier. New businesses attracted to site south of the A420 would be directed to meet the requirements of the Workplace Travel Plan, which are to be produced in consideration of Swindon Council's i-trace format.

6.3 It is therefore, proposed that the TPF is delivered, guided, coordinated and funded via establishment of a development wide Travel plan Steering Group. For the employment element of the site it is proposed to set up a Transport Management Group (TMG) as part of any subsequent phase. The steering group is to be established by first occupation with the TMG established by first occupation of an employment facility south of the A420.

6.4 This chapter of the TPF provides an introduction to what the EVSG and the TMG are, the principles of how they would operate at the EV and the benefits which they will offer over more traditional means of plan management, implementation and coordination.

6.5 The EVSG will provide a framework for the implementation of travel management initiatives that will cover all the land uses on site. The EVSG will be led by a site wide travel coordinator (SWTPC).

6.6 The TMG would be a private, non-profit making organisation. Every occupier of the employment uses south of the A420 within the site would be expected to become a member of the TMG and thereby providing funding towards the management of the TMG through membership fees.

Resources

6.7 The viability of dedicated staff time and resources is important to the implementation of the Travel Plan. The key implementer is the site wide Travel Plan Coordinator.

6.8 Initially it is proposed that a SWTPC will be employed on a part time basis. The SWTPC would be employed to assist in the establishment and formation of the EVSG and in developing a full implementation programme and strategy. It would also be the SWTPC responsibility to build the necessary partnerships and delivery of the site wide indicatives such as the Swindon EV website.

6.9 The key tasks of the SWTPC are as follows:

- To lead and manage the development and implementation of the site wide travel plan
- To act as a central point of contact for all site occupiers and external stakeholders in relation to the Travel Plan Framework
- To implement the monitoring plan set out within the monitoring strategy
- To implement and manage the data collection procedures as set out in the monitoring strategy
- To take responsibility for raising awareness of and championing sustainable travel issues
- To promote schemes and events which encourage walking, cycling and the use of public transport along with the reduction of the use of the private car
- To represent the 'human face' of the travel plan and the EVSG - explaining its purpose and the opportunities on offer
- To build and nurture the necessary partnerships required for Travel Plan implementation and success
- To promote and market the SGTPG to wider audiences
- To deliver and/or manage, procure and/or oversee delivery of site-wide travel plan measures
- To keep abreast of developing travel plan techniques
- To regularly review and evaluate both the travel plan and the operation of the EVSG and to provide feedback to the planning and highways authorities via meetings and/or other channels
- To prepare progress reports, action plans and budgets for the travel plan aimed at best meeting its objectives and targets
- To liaise with Swindon Council Travel Plan Officer
- To liaise with the individual TPCs to ensure coordination of measures and monitoring in order to maximise effective Travel Planning.

6.10 It is anticipated that the SWTPC will be employed by the developer(s) for 10 years, after which time it is expected that the EVSG will take over the organisation of the monitoring and associated measures. By this time it is expected that the EVSG will be self-funding through membership revenues.

6.11 Given the crucial role which the SWTPC will be expected to undertake, it is essential that the right individual is recruited and that he/she has the necessary skills to perform exceptionally in his/her role. Care will be taken in working up the job description for and in recruiting the SWTPC, to ensure that a poor appointment does not threaten or undermine the travel planning strategy.

Role of the Individual Travel Plan coordinators

6.12 At the time of writing, it is expected that Individual Travel Plan Coordinator (ITPC) would be employed to support the introduction of the residential travel plans that will accompany any future reserved matters application. The TPC could be employed on a part time basis responsible for the day to day running of the Travel Plan.

6.13 The ITPC will take on a facing role and will focus on delivery of initiatives as follows:

- Distribute Travel Plan welcome packs
- To act as a central transport point of contact
- To promote travel plan initiatives and measures, ensuring all EV occupiers are fully aware of the Travel Plan and services/measures available to them
- Deliver Personal Travel Planning
- Promote use of alternative modes of travel to the private car that are appropriate to individual needs
- Develop and maintain car-share database
- Produce annual newsletter informing occupiers of Travel Plan initiatives and events
- Undertake monitoring in order to measure progress against targets
- To communicate with the EVSG to ensure coordination of measures i.e. to avoid unnecessary repetition and ensure effectiveness of all Travel Plans.

Funding

6.14 In the early years of construction of the EV, it will be necessary for the developers to 'pump prime' establishment of the EVSG. As the employment land parcels are delivered in the later stages of development, the membership of the EVSG will be increased such that the membership revenues alone will be adequate to cover the costs of supporting the EVSG.

6.15 In addition, the developer is providing a failsafe fund to cover the cost of measures required in the event of the Travel Plan not meeting the required targets. The failsafe measures are only intended to mitigate the impacts of any under achievement towards meeting targets. They should not otherwise be invoked nor constitute a penalty to the developer.

Partnerships

6.16 Travel Plans require partnerships. Key stakeholders/ partners include:

- Local planning authority – Swindon Borough Council
- Transport and Highways Authority – Swindon Borough Council
- Other transport agencies – Highways Agency
- Local transport operators (e.g. bus companies, rail operators)
- Specialist advisors
- Local transport groups and forums
- National and local travel awareness and transport organisations
- Future occupiers

Branding

6.17 In order to raise awareness from the outset the Travel Plan will be branded. Branding is required to assist in promoting the importance and benefits of travel planning and obtaining by-in to sustainable travel mode choices.

6.18 The brand should be designed to be popular and instantly recognisable. It should be one of which residents and occupiers can recognise.

6.19 Marketing the Travel Plan provides an effective channel of communication between those responsible for the Travel Plan and those who benefit from the implementation of the Travel Plan.

6.20 The information will be in varied formats and able to reach all residents.. Individual Travel Plans will highlight the marketing materials available. The marketing for the site will include the following as a minimum:

- EV Travel Plan and EVSG website – providing links to valuable travel resources (e.g. journey planners, timetables etc)
- Car share website through the wider www.carshareswindon.com and dedicated car share database www.tbc.com
- Travel information / advice available from the ITPC
- Personalised Travel Planning
- Email dissemination via Travel Plan Coordinators
- Onsite marketing events, to tie into local and national promotions
- Development of high quality travel information for the development across
  Multiple formats (paper, electronic and 'interactive'/ web-based) must be complete

7 Site Wide Measures

7.1 In order to optimise the take-up of sustainable transport modes for new residents on-site, sustainable travel choices should be available and be promoted during all stages of the development process, from the design and construction, through to marketing and occupation of the Swindon EV. These measures will be applied to the current application, and will be taken forward into subsequent applications relating to south of the A420.

7.2 The focus on the mixed-use nature of the development will help to ensure that the potential number of external car driver trips is reduced compared to a solely residential development. The proposed development offers the opportunity to deliver a comprehensive mix of uses and therefore allow residents to meet a range of their day to day needs without needing to travel outside the site.

7.3 In order to achieve the objectives set out in this Travel Plan Framework, it is essential that for a development of this size, that the measures and methods used are implemented using an approach that covers the various stages outlined and divided into key phases:-

- Pre-occupation
- During occupation

7.4 The measures to be implemented during each of the three phases are described in the following sections.

Pre Occupation Phase

- A fully permeable development layout for pedestrians and cyclists reflecting desire lines– pedestrian and cycle connections will be provided from the site to link into the existing infrastructure.
- On site development streets and junctions designed to the standards outlined in the Department for Transport's Manual for Streets, to limit the dominance of the car
- Careful street design to encourage walking and cycling within the site
- Shared surfaces on residential links to promote pedestrian and cycling movements, creating some areas with a 'Home Zone' style atmosphere
- Car and cycle parking provision will be in compliance with the current guidance.
- Provision of green space for social exchange and recreation
- Appointment of a Site Wide Travel Plan Coordinator (as discussed in later sections)
- Development of 'Household Welcome Travel Packs' containing information on non-car transport modes, including public rights of way information, to be issued to each dwelling on first occupation
- Notice boards within the show house and within the work place/office to show key travel information

7.5 Through the planning process improvements to the existing public transport will be agreed. This will ensure a high quality services is provided into the development to maximize public transport usage. This will be secured through annual contributions.

Occupation Phase

Welcome Packs

It is the intention to issue Welcome packs to each household on site. A Welcome Pack could include the following information:

- A brief explanation of the travel plan and its objectives
- Contact details for the site wide Travel Plan Coordinator
- Site Wide Travel Plan website access details
- Bus Maps
- Details of any resident travel user groups including bus buddy scheme
- Cycling and Walking Maps
- Details of car share database
- Details of the Car Club
- Residents invited to register their interest in
- Other appropriate information

A dedicated travel website will be established that will include all relevant travel information, similar to that contained within the Welcome Pack, and will be review monthly and updated as required. The website will provide details on any forthcoming events and highlight the latest revisions to the TP, for example this would include the details of the rail station location and programme of delivery.

Draft Welcome Packs will be issued to the Travel Plan Officer of Swindon Council for review prior to formal issue.

Sales and Marketing office

Any websites established by the house builders on site will be encouraged to include travel information and details of the travel plan. The sales office will hold copies of the Welcome Pack and sales staff will be trained by the travel plan coordinator, if requested, as and when the sales offices are established so that they understand the principles of travel planning.

Travel Induction Sessions

The Travel Plan Coordinator will arrange travel planning sessions on a six monthly basis and invite all new residents via mail drop or email, with the aim of introducing the travel plan, the objectives and targets and to encourage discussion on the sustainable travel options. Details of any arranged sessions will be posted onto the dedicated travel web-site.

Bus Buddy Scheme

The Travel Plan Coordinator will establish a bus buddy database of those people who regular travel by public transport. Details of the scheme are to be included within the Welcome Pack and on the travel website and residents will be invited to join during the travel induction sessions.

Bicycle User Group

The Travel Plan Coordinator will establish a bicycle user group, details of which are to be included within the Welcome Pack and on the travel website and residents will be invited to join. This will be held every other month and local cycle repair shops will be invited to attend to provide advice on basic cycle repair and maintenance and to highlight what equipment is available to make cycling a pleasant experience. Prize draws are to take place at the meetings to encourage support and attendance giving users a chance to win cycling equipment.

Once the user group is set up and established, the management and organisation of the user group is to be passed to the residents with the travel plan coordinator invited to attend as and when required. The group will be supported by an annual budget of £250.

Car Share Database

The Travel Plan Coordinator will establish a dedicated Eastern Village car share database through the national company www.liftshare.com. Residents will be encouraged to join up to the database. A Prize draw will take place for those residents signing up to the databases.

A license agreement will be set up with Liftshare to initially establish the database and to carry out the annual maintenance. The website will be free to join for residents with the license subscription charge to be payable by the consortium.

Car Club

The developer will pay a £10,000 contribution per every 1000 units prior to occupation of the 500th dwelling of each 1000th dwelling towards the establishment of a car club to be used by the residents, where they will need to pay a monthly charge. It is envisaged that the contribution will be sufficient to cover the cost of the first car with any remaining funds to be used to provide incentives, which could include prize draws that offer free monthly memberships.

The site layout will include car parking spaces that are reserved for car club use within each of the neighbourhood/local centres.

Updates to travel information.

Regular updated travel information will be provided by the Travel Plan Coordinator. This will include revised copies of bus timetables as and when new timetables are published which will be posted on the travel website. The TPC will ensure that any new developments on site are effectively communicated to residents. Dedicated travel planning groups will be set up by the Travel Plan Coordinator on social networking sites like www.facebook.com. This would also be used to highlight any updates to the TP and the latest news.

Promotional events

Promotional leaflets will be issued to home owners, together with updates to the travel website and social networking groups, highlight any arranged travel campaigns. Travel campaigns will be run on an annual basis to coincide with any events being promoted by Swindon Council, events could include:

- Walk your child to school week
- Share a journey to work day
- Cycle week
- Car Club promotion week

7.6 The figure below outlines the measures to be implemented to achieve the objectives, and who would be responsible for their implementation and monitoring.

Measure	*Responsible Person*	*Timescale*
Pre-occupation Stage		
A fully permeable layout through the site for pedestrians and cyclists reflecting desire lines	Developer	Included with the approved site layout
Parking in line with current guidance	Developer	Included with the approved site layout
On-site streets and junctions designed to the standards outlined in the Department for Transport's Manual for Streets	Developer	Included with the approved site layout
Shared surfaces on residential loop lanes to promote pedestrian movements, creating a 'Home Zone' style street environment	Developer	Included with the approved site layout
Walking and cycling on site linked to existing external routes.	Developer	Included with the approved site layout
Careful street design to restrict vehicular speeds to 20mph or less within home zones	Developer	Included with the approved site layout
Secure and conveniently located cycle storage provision	Developer	Included with the approved site layout
Appointment of Travel Plan Coordinator	Developer	Prior to first occupation
Off site Infrastructure Improvements	Developer	To be delivered as per the S106
Development of Welcome Packs	Travel Plan Coordinator	Prior to first occupation
Establish steering group	Developer	Prior to first occupation
During Occupation		
Improvements to public transport routes Establish TMA	Developer Developer	On first occupation On first occupation
Distribution of Welcome Packs	Travel Plan Coordinator	On first occupation
Sales staff training	Travel Plan Coordinator	On establishment of sales office
Establish Bus Buddy databases	Travel Plan Coordinator	Prior to first occupation
Establish Cycle user group	Travel Plan Coordinator	Prior to first occupation
Provision of regular travel information	Travel Plan Coordinator	As and when necessary
Establish car club	Travel Plan Coordinator	On 500th occupation
Promotion and awareness raising packages	Travel Plan Coordinator	On an annual basis
Establish car share database	Travel Plan Coordinator	On first occupation
Establish dedicated web site	Travel Plan Coordinator	On first occupation
Coordination of appropriate reviews and monitoring of the success of the TP.	Travel Plan Coordinator	On occupation of 800th dwelling
Initial Travel Pattern Survey	Travel Plan Coordinator	On occupation of 800th dwelling
Monitoring surveys	Travel Plan Coordinator	On every 800th completion intervals
Provision of regular updated travel information	Travel Plan Coordinator	As and when necessary
Updates to the travel plan	Travel Plan Coordinator	annually
Coordination with Local Authority	Travel Plan Coordinator/CCC	Annually

8 Monitoring

Introduction

8.1 Monitoring is an important part of any goal oriented strategy. This allows outcomes and experiences to be understood as the implementation progresses. The results of regular survey can be used to inform adaptive learning during travel plan implementation. Equally, the surveys are one opportunity for further raising the awareness of the Travel Plan and sustainable travel initiatives.

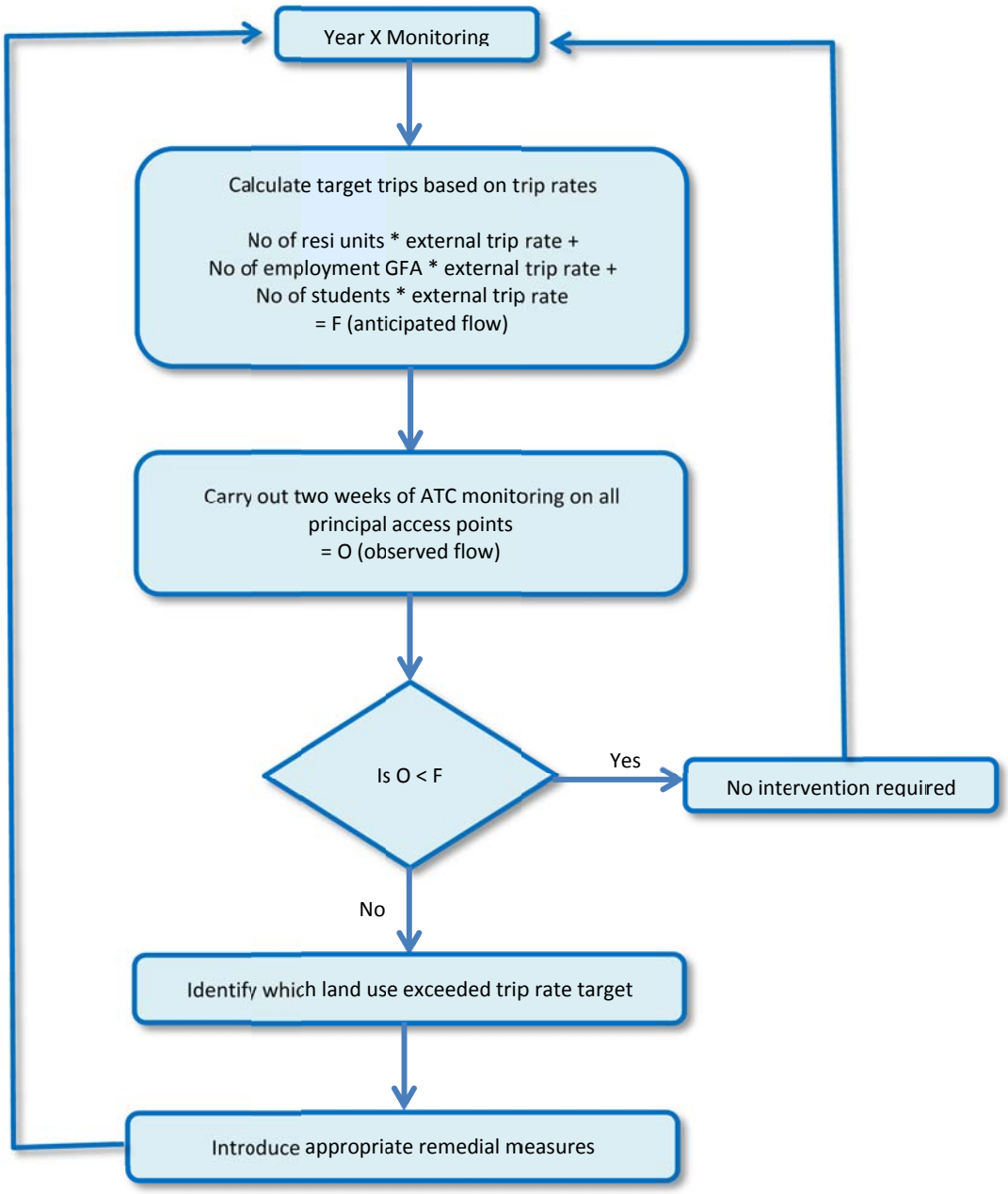
8.2 Monitoring offers the evidence base to inform travel plan review and evaluation and thereby to justify amendments that are prudent as a result of changing circumstances, emerging best practice, new innovations and technologies.

Trigger Points

8.3 Monitoring will be carried out at certain trigger points. These have been assumed to be on occupation of the 800th dwelling, then on the occupation of each additional 800th dwelling. On completion of the final dwelling, further two years of annual monitoring is to occur.

Methodology

8.4 Planning, commissioning and overseeing monitoring to be undertaken by the steering group, to be controlled by the site wide travel plan coordinator. A detailed monitoring strategy to be submitted and agreed with SBC and the Highways Agency whilst a possible monitoring methodology is shown in the flow chart below.



Remedial Measures

8.5 Should the recorded traffic levels exceed those anticipated, as set out in the subsequent paragraphs, remedial measures for implementation will be considered. The actual measures to be introduced will be dependent on which land use is identified to exceed the target trips. Measures that could be considered, include:

- Bring forward funding for car club
- Bring forward funding public transport improvements
- Personalised travel planning
- Additional promotional events

Targets

8.6 The following paragraphs set out the estimated trip generation for the proposed development at full occupation without a Travel Plan in place. Full details of vehicular trip generation methodology and trip rates by land use can be found in the Transport Assessment.

8.7 The total external two-way trips at full during the AM and PM peak hours can be broken down by mode based on Census statistics, as indicted below. This identifies the single occupancy trip to be 58.5% of the total.

Journey Type	Percentage
Car driver	66.3%
Car passenger	7.8%
Motorcycle	2.7%
Cycle	10.3%
Walk	8.0%
Bus	3.9%
Rail	0.1%
Other	0.8%

Figure 8a: Modal split

Journey Type	AM PEAK		PM Peak	
	IN	OUT	In	Out
Car driver	121	608	648	324
Car passenger	14	72	76	38
Motorcycle	5	25	26	13
Cycle	19	94	101	39
Walk	15	73	78	39
Bus	7	36	38	19
Rail	0	1	1	0
Other	1	0	8	4

Figure 8b: External trips by mode

8.8 The number of single occupancy vehicle trips (SOV) is identified below.

Journey Type	AM PEAK		PM Peak	
	IN	OUT	In	Out
Car driver	121	608	648	324
Car passenger	14	72	76	38
Single Occupancy trips	**107**	**536**	**572**	**286**

Figure 8b: Single Occupancy trips

8.9 The target of this travel plan is to reduce SOV by 10%, as shown below.

Journey Type	AM PEAK		PM Peak	
	IN	OUT	In	Out
Single Occupancy trips	107	536	572	286
Trip reduction target	11	54	57	29
Resultant External trips	**96**	**482**	**515**	**257**

Figure 8c: Single Occupancy trip reduction target